STARTUP ROADMAP GUIDE

CampusCompass

Navigate. Discover. Connect.

Software Requirements Specification (SRS)
& Complete Database Design (ERD Blueprint)

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Hatua ya Mradi (Stage):	Phase 1 & Phase 2 (Pre-Coding Blueprint)
Toleo (Version):	1.1 (Startup Edition)
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1. Utangulizi & Mtazamo wa Startup

CampusCompass si programu ya kawaida ya ramani; ni mfumo kamili wa kidigitali ulioundwa kimkakati ili kutatua changamoto ya uelekeo na mawasiliano ya ndani (Indoor/Outdoor Navigation & Campus Services Directory) kwa taasisi za elimu ya juu. Lengo letu kuu kama startup ni kuanzia na **Mbeya University of Science and Technology (MUST)** kama mradi wa majaribio (Pilot Project), na kisha kutanua huduma hii kama mfumo unaoweza kukodishwa kwa njia ya Software-as-a-Service (SaaS) kwa vyuo vingine kote nchini Tanzania na ukanda wa Afrika Mashariki.

> **Mkakati Wetu wa Kidigitali (Slogan):**
> *Navigate* — Msaidie mtumiaji kufika popote kwa urahisi.
> *Discover* — Wezesha mtumiaji kugundua huduma, ofisi, na rasilimali mpya za chuo.
> *Connect* — Unganisha wanafunzi, wafanyakazi na uongozi kupitia matangazo ya papo hapo na usaidizi wa AI.

1.1 Changamoto Tunazotatua (Problem Statement)

- **Wanafunzi Wapya Kupotea:** Kila muhula mpya, mamia ya wanafunzi wapya hupoteza muda mwingi wakitafuta madarasa (lecture rooms) au maabara.
- **Usumbufu kwa Wageni:** Wageni wanaokuja kwa ajili ya huduma za dharura au kiofisi hushindwa kupata ofisi za ma-Dean, wahadhiri au idara husika bila kuuliza watu njiani.
- **Mawasiliano Duni ya Ndani:** Hakuna chaneli moja ya kidigitali inayowafikia wanafunzi wote papo hapo wanapobadilishiwa madarasa au dharura inapotokea.

1.2 Malengo ya Mfumo (Objectives)

Lengo Kuu: Kutengeneza mfumo thabiti wa kidigitali unaounganisha teknolojia ya GPS, OpenStreetMap, na Artificial Intelligence (AI) ili kurahisisha uratibu wa maeneo na utatuzi wa changamoto za kiutawala chuoni.

Malengo Maalum:

1. Kuonyesha ramani shirikishi (interactive map) yenye majengo na kumbi zote za chuo.
2. Kupata alama halisi ya GPS (user's real-time location) na kuhesabu njia fupi zaidi (optimal route calculation).
3. Kutoa maelekezo ya hatua kwa hatua ya kutembea kwa miguu (step-by-step walking directions).
4. Kuwezesha mifumo ya AI kusaidia wanafunzi kujibu maswali ya papo hapo kuhusu chuo.

2. Mfumo wa Teknolojia & High-Level Architecture

Ili kuhakikisha mfumo una uwezo wa kukua (scaling) kutoka chuo kimoja (MUST) hadi vyuo mamia barani Afrika, tumechagua teknolojia za kisasa ambazo ni salama, za haraka (highly performant) na zenye gharama nafuu za uendeshaji:

Sehemu ya Mfumo	Teknolojia Iliyoteuliwa	Sababu ya Kiufundi (Startup Advantage)
Backend	Python, FastAPI, SQLAlchemy	Kasi kubwa sana ya API response, asynchronicity (async/await), na autogenerated Swagger docs kwa haraka ya ujenzi.
Database	PostgreSQL, PostGIS	Uwezo wa juu wa SQL na PostGIS ugani wa kuhifadhi na kupigia hesabu data za kijiografia (Coordinates, GPS Points).
Mobile & Web	Flutter (Dart) / React.js	Flutter inaruhusu kodi moja kutumika Android na iOS, ikipunguza nusu ya gharama na muda wa startup development.
AI Assistant	OpenAI API & Python ML	Inatumia Retrieval-Augmented Generation (RAG) ili AI ijibu maswali maalum kulingana na muundo wa chuo.
Deployment	Docker, AWS, Nginx	Kufunga mfumo kwenye "Containers" kunarahisisha kuuhamisha au kuuweka kwenye vyuo vingine kwa sekunde chache.

Kumbuka: Katika awamu hii ya pili (Phase 2), tutazama moja kwa moja kwenye **Database Design** ili kutengeneza database schema ya kisasa inayoweza kuhimili vyuo vingi kwa wakati mmoja (Multi-Tenant Architecture).

3. Usanifu wa Database (Phase 2: Database Design)

Huu ndio moyo wa mfumo wetu. Tumeunda database schema yenye meza (tables) 11 zilizounganishwa kimantiki kuzuia upotevu wa data (normalization) na kurahisisha maswali ya haraka ya utafutaji (indexing & queries).

3.1 Orodha ya Meza na Uhusiano wake

1. Table: `campuses`

Inahifadhi taarifa za chuo husika (Mfano: MUST Main Campus, UDSM Mlimani). Hii inafanya programu kuwa ya Multi-tenant SaaS.

Field Name	Data Type	Key / Constraint	Description
id	UUID / INT	PRIMARY KEY	Unique identifier ya chuo.
name	VARCHAR(150)	NOT NULL, UNIQUE	Jina la chuo (e.g., "MUST").
location_city	VARCHAR(100)	NOT NULL	Mji uliopo chuo (e.g., "Mbeya").
latitude	DOUBLE PRECISION	NOT NULL	Center GPS latitude.
longitude	DOUBLE PRECISION	NOT NULL	Center GPS longitude.

2. Table: `users`

Inahifadhi taarifa kuu za watumiaji wote wa mfumo bila kujali nafasi zao.

Field Name	Data Type	Key / Constraint	Description
id	UUID	PRIMARY KEY	Auto-generated UUID.
campus_id	UUID	FOREIGN KEY	Inarejelea chuo mtumiaji alipo.
email	VARCHAR(120)	UNIQUE, NOT NULL	Barua pepe kwa ajili ya login.
password_hash	VARCHAR(255)	NOT NULL	Nenosiri lililofungwa kwa bcrypt hashing.
role	VARCHAR(20)	NOT NULL	Mihimili: ADMIN, STUDENT, STAFF, VISITOR.
is_active	BOOLEAN	DEFAULT TRUE	Kujua kama akaunti ipo hai au imefungwa.

3. Table: `students`

Inahifadhi taarifa za ziada za wanafunzi tu (Ina uhusiano wa 1-to-1 na `users`).

Field Name	Data Type	Key / Constraint	Description
user_id	UUID	PK , FK	Inarejelea meza ya `users.id`.
reg_number	VARCHAR(50)	UNIQUE, NOT NULL	Namba ya usajili ya chuo (e.g., "21100523001").
department	VARCHAR(100)	NOT NULL	Idara ya mwanafunzi (e.g., "CoICT").

4. Table: `staff`

Inahifadhi taarifa za walimu au wafanyakazi (Ina uhusiano wa 1-to-1 na `users`).

Field Name	Data Type	Key / Constraint	Description
user_id	UUID	PK , FK	Inarejelea meza ya `users.id`.
staff_id_number	VARCHAR(50)	UNIQUE, NOT NULL	Nambari ya ajira ya mfanyakazi.
designation	VARCHAR(100)	NOT NULL	Cheo chake (e.g., "Senior Lecturer").

5. Table: `buildings`

Majengo yaliyopo kwenye campus (Ina uhusiano wa 1-to-Many na `campuses`).

Field Name	Data Type	Key / Constraint	Description
id	UUID	PRIMARY KEY	Unique ID ya jengo.
campus_id	UUID	FOREIGN KEY	Chuo ambalo jengo hili lipo.
name	VARCHAR(150)	NOT NULL	Jina la jengo (e.g., "Block B - Engineering").
description	TEXT	NULLABLE	Maelezo ya jumla kuhusu jengo hili.
latitude	DOUBLE PRECISION	NOT NULL	GPS Latitude point ya kuingilia jengo.
longitude	DOUBLE PRECISION	NOT NULL	GPS Longitude point ya kuingilia jengo.

6. Table: `rooms`

Kumbi au madarasa ndani ya jengo maalum (Ina uhusiano wa 1-to-Many na `buildings`).

Field Name	Data Type	Key / Constraint	Description
id	UUID	PRIMARY KEY	Unique identifier.
building_id	UUID	FOREIGN KEY	Jengo lililopo darasa hili.
room_number	VARCHAR(50)	NOT NULL	Nambari au jina la chumba (e.g., "Room E101").
floor	INT	NOT NULL DEFAULT 0	Ghorofa lilipo (0 = Ground, 1 = First Floor, nk).
capacity	INT	NULLABLE	Idadi ya juu ya wanafunzi wanaotoshea.

4. SQL Database Schema & ERD Text Representation

Kama timu ya maendeleo, kuelewa jinsi gani meza hizi zinavyoungana katika muundo wa SQL (Data Definition Language) kunatusaidia kuandika API bila makosa ya uingizaji data. Hapa chini kuna muundo thabiti wa SQL kwa ajili ya PostgreSQL:

```
-- 1. Create Campuses Table
CREATE TABLE campuses (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name VARCHAR(150) NOT NULL UNIQUE,
    location_city VARCHAR(100) NOT NULL,
    latitude DOUBLE PRECISION NOT NULL,
    longitude DOUBLE PRECISION NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Create Users Table
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    campus_id UUID NOT NULL REFERENCES campuses(id) ON DELETE CASCADE,
    email VARCHAR(120) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) NOT NULL CHECK (role IN ('ADMIN', 'STUDENT', 'STAFF', 'VISITOR')),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Create Students Table (1-to-1 with Users)
CREATE TABLE students (
    user_id UUID PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE,
    reg_number VARCHAR(50) NOT NULL UNIQUE,
    department VARCHAR(100) NOT NULL
);

-- 4. Create Staff Table (1-to-1 with Users)
CREATE TABLE staff (
    user_id UUID PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE,
    staff_id_number VARCHAR(50) NOT NULL UNIQUE,
    designation VARCHAR(100) NOT NULL
);

-- 5. Create Buildings Table
CREATE TABLE buildings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    campus_id UUID NOT NULL REFERENCES campuses(id) ON DELETE CASCADE,
    name VARCHAR(150) NOT NULL,
    description TEXT,
    latitude DOUBLE PRECISION NOT NULL,
    longitude DOUBLE PRECISION NOT NULL
```

```
);

-- 6. Create Rooms Table
CREATE TABLE rooms (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    building_id UUID NOT NULL REFERENCES buildings(id) ON DELETE CASCADE,
    room_number VARCHAR(50) NOT NULL,
    floor INT NOT NULL DEFAULT 0,
    capacity INT
);
```

Siri ya Kufanikiwa kwa Startup (Index Tuning):
Ili database isichelewe (latency chini ya 2s) chuo kinapokuwa na maelfu ya watumiaji kwa sekunde moja, tutaweka faharasa (database indices) kwenye kigezo cha `latitude` na `longitude` pamoja na `email` ili kufanya database searches kuwa nyepesi mno.

5. Upangaji wa Njia za Mtandao (API Paths) & Deployment

Kupitia mfumo wa FastAPI, tunaunganisha meza zetu zote kuwa mwisho wa njia (Endpoints) ambazo programu ya simu (Flutter mobile) au kompyuta (React web dashboard) zinaweza kuziita kuonesha taarifa.

Orodha ya Mwisho wa Njia (Core API Endpoints):

- `POST /api/v1/auth/register` — Kusajili watumiaji wapya (Wanafunzi/Wageni).
- `POST /api/v1/auth/login` — Kupata token maalum (JWT) ili kufikia huduma za siri.
- `GET /api/v1/campuses/{id}/buildings` — Kuorodhesha majengo yote ya chuo maalum.
- `GET /api/v1/buildings/{id}/rooms` — Kupata vyumba na madarasa ya jengo husika.
- `POST /api/v1/navigation/route` — Kutuma pointi mbili za coordinates na kupokea njia ya hatua kwa hatua kwa kutumia hesabu ya ramani.

CampusCompass: Kazi Imeanza Rasmi!

Huu ndio mwongozo wa msingi utakaomwezesha kila mhandisi kuandika mifumo kulingana na ramani hii. Hatua inayofuata sasa ni kuanza ujenzi halisi wa msimbo (FastAPI backend skeleton).